



Week 11 - Graded Assignment 11



The due date for submitting this assignment has passed.

Due on 2026-04-29, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-29, 11:51 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

1 point

Consider the following function

$$f(x, y, z) = x^2 z + y^3 e^{xy} + \sin(xyz). f(x, y, z) = x^2 z + y^3 e^{xy} + \sin(xyz).$$

Which of the following option(s) is (are) true?

☐

$$f_{xx} = 2z + y^5 e^{xy} - yz \sin(xyz) f_{xx} = 2z + y^5 e^{xy} - yz \sin(xyz)$$

☒

$$f_{xx} = 2z + y^5 e^{xy} - (yz)^2 \sin(xyz) f_{xx} = 2z + y^5 e^{xy} - (yz)^2 \sin(xyz)$$

☐

$$f_{zy} = \cos(xyz) x^2 y z f_{zy} = \cos(xyz) x^2 y z$$

☐

$$f_{xz} = 2x - x y^2 z \sin(xyz) f_{xz} = 2x - x y^2 z \sin(xyz)$$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$f_{xx} = 2z + y^5 e^{xy} - (yz)^2 \sin(xyz) f_{xx} = 2z + y^5 e^{xy} - (yz)^2 \sin(xyz)$$

Number of critical points of the function $f(x, y, z) = e^x (x^2 - y^2 - 2z^2)$

$$f(x, y, z) = e^x (x^2 - y^2 - 2z^2) \text{ is}$$

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Choose the correct statement(s) about $f(x, y) = \sqrt{x^2 + y^2} f(x, y) = \sqrt{x^2 + y^2}$

$\sqrt{}$

.

☐

f has no critical points.

☒

$(0, 0)$ is the only critical point of f .

☐

$(0, 0)$ is a local maximum of f .

☒

$(0, 0)$ is a local minimum of f .

☒


$(0, 0)$ is a global minimum of f .

Yes, the answer is correct.

Score: 1

Accepted Answers:

$(0, 0)$ is the only critical point of f .

$(0, 0)$ is a local minimum of f .

$(0, 0)$ is a global minimum of f .

Find the minimum value of the surface area of a rectangular box that has a volume of 1000 unit^3 .

Note: If x, y and z represent the lengths of the three sides of a rectangular box, then the surface area and volume are given by $2(xy + yz + zx)$ and xyz , respectively.

600

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 600

1 point

1 point

Consider a function $f(x, y, z) = x^3 + 3y^3 + z^3 - 4x - 6y - 4z$

$f(x, y, z) = x^3 + 3y^3 + z^3 - 4x - 6y - 4z$. If A is the Hessian matrix of $f(x, y, z)$ at $(0, 1, 1)$, then which of the following option(s) is(are) true?



$\det(A) = 0$



$\text{Rank}(A) = 2$



$\text{Nullity}(A) = 1$



$\det(A) = 648$



$\text{Rank}(A) = 3$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\det(A) = 0$

$\text{Rank}(A) = 2$

$\text{Nullity}(A) = 1$

1 point

Consider the following function

$$f(x, y) = e^x + e^y - e^{x+y} \quad f(x, y) = e^x + e^y - e^{x+y}$$

Which of the following option(s) is (are) true?



Hessian matrix of f at point $(1, 1)$ is $\begin{bmatrix} 1 - e & -e^2 \\ -e^2 & 1 - e \end{bmatrix}$.



Hessian matrix of f at point $(1, 1)$ is $\begin{bmatrix} e - e^2 & -e^2 \\ -e^2 & e - e^2 \end{bmatrix}$.



f has a point of maximum.



f has a saddle point.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Hessian matrix of f at point $(1, 1)(1, 1)$ is $\begin{bmatrix} e - e^2 & -e^2 \\ -e^2 & e - e^2 \end{bmatrix} [e - e^2 - e^2 - e^2 e - e^2]$.

f has a saddle point.

1 point

Choose the correct statements.



If $f: R^3 \rightarrow R$ then the Hessian matrix $H_f(x, y, z)$ is same for all $(x, y, z) \in R^3$.



If $f: R^3 \rightarrow R$ is a linear transformation then the Hessian matrix $H_f(x, y, z)$ is same for all $(x, y, z) \in R^3$.



For each $n \in \mathbb{N}$, there exists a map $f: R^n \rightarrow R$ such that the rank of the Hessian matrix $H_f(x_1, x_2, \dots, x_n)$ is n for some $(x_1, x_2, \dots, x_n) \in R^n$.



For each $n \in \mathbb{N}$, there exists a map $f: R^n \rightarrow R$ such that the Hessian matrix $H_f(x_1, x_2, \dots, x_n)$ is symmetric for all $(x_1, x_2, \dots, x_n) \in R^n$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If $f: R^3 \rightarrow R$ is a linear transformation then the Hessian matrix $H_f(x, y, z)$ is same for all $(x, y, z) \in R^3$.

For each $n \in \mathbb{N}$, there exists a map $f: R^n \rightarrow R$ such that the rank of the Hessian matrix $H_f(x_1, x_2, \dots, x_n)$ is n for some $(x_1, x_2, \dots, x_n) \in R^n$.

For each $n \in \mathbb{N}$, there exists a map $f: R^n \rightarrow R$ such that the Hessian matrix $H_f(x_1, x_2, \dots, x_n)$ is symmetric for all $(x_1, x_2, \dots, x_n) \in R^n$.

The maximum value of the function $-2x^2 + 2xy - y^2 + 12x - 4y - 7$ is

13

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 13

1 point

1 point

Define a function

$$f(x, y) = \begin{cases} (x - y)^2 \sin\left(\frac{1}{x - y}\right) & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

Choose the set of correct options.



f is continuous at $(0, 0)$.



$f_x(0, 0) \neq f_y(0, 0)$



$f_x(0, 0) = f_y(0, 0)$



f is differentiable at $(0, 0)$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

f is continuous at $(0, 0)$.

$f_x(0, 0) = f_y(0, 0)$

f is differentiable at $(0, 0)$.

A company produces two types of chocolates, one that sells for Rs.3 and another at Rs.22. The total revenue, in lakhs, from the sale of x lakhs chocolates at Rs.22 each and y lakhs at Rs.33 each is given by

$$F(x, y) = 2x + 3y$$

The total cost in lakhs for producing x lakhs of the Rs.22 chocolates and y lakhs of the Rs.33 chocolates is given by

$$G(x, y) = 2x^2 - 2xy + y^2 - 9x + 6y + 7$$

Then the maximum profit in lakhs is

[Enter your answer upto two decimal places]

11.25

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 11.25

1 point

1 point

(MSQ) Let $f: R^2 \rightarrow R$ be a multivariable function and $(x_0, y_0) \in R^2$.

Consider the following statements:

- P: f is continuous at (x_0, y_0) .
- Q: The partial derivatives of f exist and are continuous in a neighborhood of (x_0, y_0) .
- R: The tangent plane to the graph of f exists at (x_0, y_0) .

Choose the correct statements from the following.



P implies that f is differentiable at (x_0, y_0) .



If f is differentiable at (x_0, y_0) , then P is satisfied.



Q implies that f is differentiable at (x_0, y_0) .



If f is differentiable at (x_0, y_0) , then Q is satisfied.



R implies that f is differentiable at (x_0, y_0) .



If f is differentiable at (x_0, y_0) , then R is satisfied.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If f is differentiable at (x_0, y_0) , then P is satisfied.

Q implies that f is differentiable at (x_0, y_0) .

R implies that f is differentiable at (x_0, y_0) .

If f is differentiable at (x_0, y_0) , then R is satisfied.

(NAT) A particle moving in a potential field tends to settle at points where the potential energy is a local minimum (stable equilibrium point). Consider a particle moving in R^2 where the potential energy as a function of position is given by

$$V(x, y) = x^4 + y^4 - 4xy + 1$$

for all $(x, y) \in R^2$. Using the Hessian test, find the potential energy at stable equilibrium points.

-1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -1

1 point